

HW 3

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$$1) \begin{matrix} T_{s1} & & T_{s2} \\ \left| \begin{matrix} \dot{q} \\ \cdot \end{matrix} \right| & \left| \begin{matrix} \hat{q} \\ x_1 \end{matrix} \right| & \left| \begin{matrix} \dot{q} \\ x_2 \end{matrix} \right| & \left| \begin{matrix} \cdot \\ x_3 \end{matrix} \right| \end{matrix}$$

$$0 < x < x_1 \quad K \frac{\partial^2 T_1}{\partial x^2} = -\dot{q} = -\dot{q}_0 (1 + \alpha x) \quad 1)$$

$$x_1 < x < x_2 \quad K \frac{\partial^2 T_2}{\partial x^2} = \dot{q} = \hat{q} \quad 2)$$

$$x_2 < x < x_3 \quad K \frac{\partial^2 T_3}{\partial x^2} = -\dot{q} = -\dot{q}_0 (1 + \alpha x) \quad 3)$$

$$1) \quad T_1(x) = -\frac{\dot{q}_0}{K_1 \cdot 2} x^2 + \frac{-\dot{q}_0 \cdot \alpha}{K_1 \cdot 6} x^3 + c_1 x + c_2$$

$$2) \quad T_2(x) = \frac{\hat{q}}{K_2 \cdot 2} x^2 + c_3 x + c_4$$

$$3) \quad T_3(x) = -\frac{\dot{q}_0}{K_3 \cdot 2} x^2 + \frac{-\dot{q}_0 \cdot \alpha}{K_3 \cdot 6} x^3 + c_5 x + c_6$$

$$1) \quad -K_1 \frac{\partial T_1}{\partial x} = -K_1 \left(-\frac{\dot{q}_0}{K_1} x + \frac{-\dot{q}_0 \cdot \alpha}{K_1 \cdot 2} x^2 + c_1 \right)$$

$$2) \quad -K_2 \frac{\partial T_2}{\partial x} = -K_2 \left(\frac{\hat{q}}{K_2} x + c_3 \right)$$

$$3) \quad -K_3 \frac{\partial T_3}{\partial x} = -K_3 \left(-\frac{\dot{q}_0}{K_3} x + \frac{-\dot{q}_0 \cdot \alpha}{K_3 \cdot 2} x^2 + c_5 \right)$$

B.C

$$T_1(x_1) = T_2(x_1) \rightarrow -\frac{\dot{q}_0}{K_1 \cdot 2} x_1^2 + \frac{-\dot{q}_0 \cdot \alpha}{K_1 \cdot 6} x_1^3 + c_1 x_1 + c_2 = \frac{\hat{q}}{K_2 \cdot 2} x_1^2 + c_3 x_1 + c_4$$

$$T_3(x_2) = T_2(x_2) \rightarrow -\frac{\dot{q}_0}{K_3} x_2^2 + \frac{-\dot{q}_0 \cdot \alpha}{K_3 \cdot 6} x_2^3 + c_5 x_2 + c_6 = \frac{\hat{q}}{K_2 \cdot 2} x_2^2 + c_3 x_2 + c_4$$

$$T_1(0) = T_{s1} \rightarrow 0 + 0 + 0 + c_2 = T_{s1}$$

$$T_3(x_3) = T_{s2} \rightarrow -\frac{\dot{q}_0}{K_3 \cdot 2} x_3^2 + \frac{-\dot{q}_0 \cdot \alpha}{K_3 \cdot 6} x_3^3 + c_5 x_3 + c_6 = T_{s2}$$

$$+K_1 \frac{\partial T_1}{\partial x} \bigg|_{x_1} = K_2 \frac{\partial T_2}{\partial x} \bigg|_{x_1} \rightarrow \dot{q}_0 x_1 + \frac{\dot{q}_0 \cdot \alpha}{2} x_1^2 - K_1 c_1 = -\hat{q} x_1 + -K_2 c_3$$

$$K_3 \frac{\partial T_3}{\partial x} \bigg|_{x_2} = K_2 \frac{\partial T_2}{\partial x} \bigg|_{x_2} \rightarrow \dot{q}_0 x_1 + \frac{\dot{q}_0 \cdot \alpha}{2} x_1^2 - K_3 c_5 = -\hat{q} x_1 + -K_2 c_3$$

linear eqns to solve for

$$c_1 \rightarrow c_6$$

but lengthy makes $T(x)$ $T'(x)$

unreasonably long and hard to look at

if given values for constants

unknown $(q_0, \alpha, x_1, x_2, x_3, K_1, K_2, K_3, \hat{q}_0, \hat{q})$

$c_1 \rightarrow c_6$ can easily be calculated

as engineers that is the real

calculation way for this specific case

$$2) \begin{matrix} K_S & & K_F \\ \left| \begin{matrix} \textcircled{1} \end{matrix} \right| & \left| \begin{matrix} \dot{q} \\ \textcircled{2} \\ | \end{matrix} \right| & \left| \begin{matrix} \textcircled{3} \end{matrix} \right| & \left| \begin{matrix} T_{\infty}, \eta \end{matrix} \end{matrix}$$

$$x = -(b+L) \quad x = -L \quad x = 0 \quad x = L \quad x = b+L$$

$$2) \quad f: \quad K_F \frac{\partial^2 T_F}{\partial x^2} = -\dot{q} \quad -L < x < L$$

$$T_F = -\frac{\dot{q}}{K_F} \frac{x^2}{2} + c_1 x + c_2$$

$$1,3) \quad S_L: \quad K_S \frac{\partial^2 T_S}{\partial x^2} = 0 \quad \begin{matrix} -(b+L) < x < -L \\ L < x < b+L \end{matrix}$$

$S_R:$

$$T_{S_L} = c_3 x + c_4$$

$$T_{S_R} = c_5 x + c_6$$

B.C

$$-K_S \frac{\partial T_{S_L}}{\partial x} \bigg|_{x=-(b+L)} = 0 \rightarrow 0 = c_3 (-b-L) \quad \boxed{c_3 = 0}$$

$$-K_S \frac{\partial T_{S_R}}{\partial x} \bigg|_{x=b+L} = h(T_{S_R}(b+L) - T_{\infty}) \rightarrow \boxed{-\frac{h}{K_S} (T_{S_R}(b+L) - T_{\infty}) = c_5}$$

$$T_F(-L) = T_{S_L}(-L) \rightarrow -\frac{\dot{q}}{K_F} \frac{L^2}{2} + c_1 L + c_2 = -\cancel{c_3} L + c_4$$

$$T_F(L) = T_{S_R}(L) \rightarrow -\frac{\dot{q}}{K_F} \frac{L^2}{2} + c_1 L + c_2 = c_5 L + c_6$$

$$-K_S \frac{\partial T_{S_L}}{\partial x} \bigg|_{-L} = -K_F \frac{\partial T_F}{\partial x} \bigg|_{-L} \rightarrow +K_S (\cancel{c_3}) = -\cancel{K_F} \frac{\dot{q}}{K_F} L + c_1$$

$$\boxed{c_1 = -\frac{\dot{q} L}{K_S}}$$

$$-K_S \frac{\partial T_{S_R}}{\partial x} \bigg|_L = -K_F \frac{\partial T_F}{\partial x} \bigg|_L \rightarrow -K_S (c_5) = +\cancel{K_F} \frac{\dot{q}}{K_F} L + c_1$$

$$\boxed{c_5 = -\frac{2L\dot{q}}{K_S}}$$

linear poly linear